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| COURSE NAME: Software Engineering       | COURSE NUMBER: CS3151 | SECTION: 1, 2, 3 |

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### Lab Objectives:

- Identify target users for the software project
- Develop realistic user personas
- Map personas to software requirements
- Apply user-centered design thinking

### Part 1: Identify User Groups

Task: List all potential user groups of the PlayGCC system.

### Deliverable:

| User Group                      | Role                                                                                                   | Frequency of Use   | Technical Level |
|---------------------------------|--------------------------------------------------------------------------------------------------------|--------------------|-----------------|
| Casual / Social Players         | Individual padel, tennis, football & swimming players who book courts and look for people to play with | Daily–Weekly       | Medium          |
| Recurring Teams & Padel Pairs   | Fixed groups that book the same time slot every week and occasionally need a substitute player         | Weekly             | Medium          |
| Women & Family Players          | Players seeking women-only or family-friendly sessions                                                 | Weekly–Occasional  | Low–Medium      |
| Venue Owners / Managers         | Padel, tennis, football & swimming venues listing courts and managing bookings, pricing and demand     | Daily              | Medium–High     |
| Corporate Wellness Coordinators | HR/office admins booking recurring slots for employee wellness programs                                | Monthly–Occasional | Medium          |

### Part 2: Persona Creation

Three personas were created, covering PlayGCC's core consumer segments (a casual/social player and a recurring group organizer) and its B2B segment (a venue manager).



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| <b>Persona Name:</b>          | <b>Sara Al-Otaibi — The Social Player</b>                                                                                                                                                                                                        |
| <b>Age:</b>                   | 27                                                                                                                                                                                                                                               |
| <b>Education:</b>             | Bachelor's degree in Marketing                                                                                                                                                                                                                   |
| <b>Role:</b>                  | Marketing Specialist at a Riyadh-based firm; plays padel and swims casually 2–3 times a week                                                                                                                                                     |
| <b>Goals:</b>                 | Play padel or go swimming regularly without relying on one fixed group of friends<br>Feel safe playing with people she hasn't met before                                                                                                         |
| <b>Motivations:</b>           | Staying active and social outside of work<br>Enjoys the community and competitive side of casual sport                                                                                                                                           |
| <b>Pain Points:</b>           | Her regular padel partner often cancels last-minute, leaving her without a game<br>Uncomfortable joining mixed-gender open games with strangers of unknown skill or intent<br>Currently relies on scattered WhatsApp groups to find a 4th player |
| <b>Technical Skill Level:</b> | Medium — comfortable with apps, uses 3–4 booking/social apps regularly                                                                                                                                                                           |
| <b>Usage Context:</b>         | Books mostly from her phone, in the evening after work, often on short notice (within 1–2 hours of playing)                                                                                                                                      |

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| <b>Persona Name:</b>          | <b>Faisal Al-Harbi — The Recurring Group Organizer</b>                                                                                                                                                        |
| <b>Age:</b>                   | 34                                                                                                                                                                                                            |
| <b>Education:</b>             | Bachelor's degree in Information Technology                                                                                                                                                                   |
| <b>Role:</b>                  | IT Consultant; organizes a weekly 5-a-side football match with colleagues and friends                                                                                                                         |
| <b>Goals:</b>                 | Lock in the same weekly slot without re-booking it manually every time<br>Split the cost fairly among 8–10 players without chasing anyone for money                                                           |
| <b>Motivations:</b>           | Being the reliable one who keeps the weekly game running<br>Avoiding the admin overhead of coordinating a large group                                                                                         |
| <b>Pain Points:</b>           | Re-books and re-messages the same group every single week<br>Ends up fronting the full court cost and collecting cash/transfers afterward<br>Loses the group's preferred slot when he books a little too late |
| <b>Technical Skill Level:</b> | Medium-High — power user of WhatsApp groups, comfortable managing bookings digitally                                                                                                                          |
| <b>Usage Context:</b>         | Books from his phone once a week, usually a few days in advance; occasionally books on desktop when planning a full month                                                                                     |

|                      |                                                                                                                                                                 |
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| <b>Persona Name:</b> | <b>Munira Al-Qahtani — The Venue Manager</b>                                                                                                                    |
| <b>Age:</b>          | 42                                                                                                                                                              |
| <b>Education:</b>    | Bachelor's degree in Business Administration                                                                                                                    |
| <b>Role:</b>         | Manager of a mid-size padel & tennis club in Riyadh with 6 courts                                                                                               |
| <b>Goals:</b>        | Keep courts booked at all hours, not just evenings and weekends<br>Replace her paper/phone booking process with something her front-desk staff can use directly |



|                               |                                                                                                                                                                                                                                 |
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| <b>Motivations:</b>           | Growing the club's revenue and reputation<br>Reducing the time her staff spend on the phone taking bookings                                                                                                                     |
| <b>Pain Points:</b>           | Weekday afternoon courts (2–4pm) sit empty most days<br>No visibility into demand patterns to know which slots to discount<br>Currently manages bookings via phone calls and a shared spreadsheet, which causes double-bookings |
| <b>Technical Skill Level:</b> | Medium — uses email, spreadsheets and basic business software daily, but not a technical user                                                                                                                                   |
| <b>Usage Context:</b>         | Uses a desktop dashboard at the front desk during the day, and checks a mobile summary from home in the evening                                                                                                                 |

### Part 3: Persona to Requirements Mapping

Task: Map each persona to the PlayGCC features that serve their goals and pain points.

| Persona                  | Feature Needed                                                | Priority |
|--------------------------|---------------------------------------------------------------|----------|
| Sara (Social Player)     | Quick Match — find a verified player nearby within hours      | High     |
| Sara (Social Player)     | Verified profiles & two-way post-match ratings                | High     |
| Sara (Social Player)     | Women-only / family session filter                            | High     |
| Sara (Social Player)     | Unified multi-venue booking view                              | Medium   |
| Sara (Social Player)     | Split payment on booking (Mada/STC Pay/Apple Pay)             | Medium   |
| Sara (Social Player)     | AI profile summarization                                      | Low      |
| Faisal (Group Organizer) | Recurring bookings ("every Tuesday 8pm")                      | High     |
| Faisal (Group Organizer) | Split payment on booking, auto-divided among players          | High     |
| Faisal (Group Organizer) | Waitlist auto-notify for full prime slots                     | Medium   |
| Faisal (Group Organizer) | Recurring group + stranger hybrid ("need a 4th", auto-posted) | Medium   |
| Faisal (Group Organizer) | WhatsApp booking bot                                          | Medium   |
| Faisal (Group Organizer) | Off-peak dynamic pricing deals                                | Low      |
| Munira (Venue Manager)   | B2B booking dashboard / white-label tools                     | High     |
| Munira (Venue Manager)   | Demand forecasting for venues                                 | High     |
| Munira (Venue Manager)   | Off-peak dynamic pricing controls                             | High     |
| Munira (Venue Manager)   | Real-time availability sync across the marketplace            | Medium   |
| Munira (Venue Manager)   | Corporate wellness booking management                         | Medium   |
| Munira (Venue Manager)   | Nafath-verified customer visibility for staff                 | Low      |



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### Part 4: Scenarios & User Stories

#### Sara Al-Otaibi — The Social Player

##### Usage Scenario:

Sara finishes work and wants to play padel, but her regular partner cancels last-minute. She opens PlayGCC, taps Quick Match, and filters for a women-only session within 5 km. Within ten minutes she is matched with a verified intermediate player, books the court, pays her half through split payment, and gets a reminder an hour before the match.

##### User Stories:

- As a social player, I want to find a verified player nearby within minutes so that I don't have to cancel my game when my usual partner can't make it.
- As a female player, I want to filter for women-only sessions so that I can play in a setting I feel comfortable in.

#### Faisal Al-Harbi — The Recurring Group Organizer

##### Usage Scenario:

Faisal organizes a weekly football match with eight colleagues every Tuesday at 8pm. Instead of re-booking and re-collecting payment every week, he sets up a recurring booking in PlayGCC. Each week the app automatically reserves the pitch, sends a split-payment request to the group, and notifies Faisal if the venue is fully booked so he can join a waitlist for a better slot.

##### User Stories:

- As a recurring group organizer, I want to set up an automatic weekly booking so that I don't have to manually reserve and message everyone every week.
- As a group organizer, I want the cost split and auto-charged to each player so that I'm not stuck fronting the bill.

#### Munira Al-Qahtani — The Venue Manager

##### Usage Scenario:

Munira manages a padel club whose courts sit empty most weekday afternoons. Using PlayGCC's venue dashboard, she reviews a demand forecast that flags 2–4pm as consistently underbooked, sets a 20% off-peak discount for that window, and watches the deal get pushed to nearby price-sensitive players automatically. Within a week her afternoon utilization improves, and her front-desk staff manage all bookings from the same dashboard instead of a shared spreadsheet.

##### User Stories:

- As a venue manager, I want to see demand forecasts for my courts so that I can set targeted discounts for slow time slots.
- As a venue manager, I want a real-time booking dashboard so that my staff no longer manage reservations by phone or spreadsheet.



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**Submission:**

- Personas (3)
- Persona-feature mapping
- User stories (1 scenario + 2 user stories per persona)

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**Assessment Criteria:**

| Criterion                   | Marks |
|-----------------------------|-------|
| Persona realism & clarity   | 30%   |
| Alignment with project idea | 30%   |
| Requirements mapping        | 20%   |
| Scenarios & user stories    | 20%   |